

are utilised in the travel industry for tasks like booking flights, hotels, or rental cars. They can also offer travel recommendations and information about destinations and answer frequently asked questions. Chatbots can assist in itinerary management and provide real-time updates about travel schedules. In the insurance industry, chatbots help customers file claims,

obtain policy information, and answer questions about coverage. They can guide users through claims, collect relevant information, and offer instant assistance. Table 1 shows examples of how different industries are leveraging AI-based chatbots for their businesses.

Several AI-based chatbot tools are available in the market, each with its own features and capabilities (Table 2).

Using IBM Watson to develop a chatbot application

The steps we need to follow when using IBM Watson to deploy an AI-based chatbot solution are briefly listed below.

Step 1. Set up an IBM Cloud account: If you don't already have an IBM Cloud account, sign up at <https://cloud.ibm.com/registration>.

Table 2: Comparison of popular AI-based chatbots

Criteria	WP-Chatbot	Microsoft Bot	IBM Watson Assistant	CSML	LivePerson
Developed by	MobileMonkey	Microsoft	IBM	Clevy.io	LivePerson, Inc.
AI technology used	NLP	NLP	NLP, ML	NLP, ML	NLP
Key features	Facebook page branding, compatible with WordPress	Allows use of open source SDK	Provides a visual interface for designing chatbot flows, and supports multiple channels	Provides a ready-to-use chatbot templates library	Hyperlinks and custom pre-written statements
Target application	WordPress ecosystem	To build a conversational AI experience	To respond to user queries for commerce and support	To improve the customer experience for e-commerce platforms	To engage customers using messaging, voice, etc
Pricing	14-day free trial; and US\$ 99/month	No free trial US\$ 499/month	US\$ 140/instance	Free version available	Free trial available

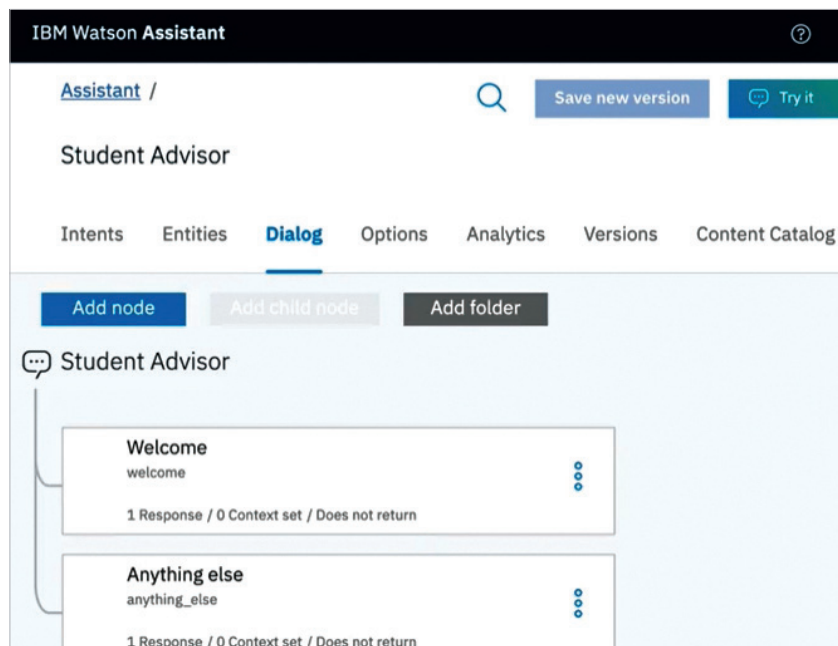


Figure 2: Create a chatbot assistant to address customer queries

Step 2. Create an IBM Watson Assistant service: Once you successfully log in at IBM Watson, create an assistant service using the catalogue menu bar. Now, generate assistant and dialogue skills. Give a name to your dialogue skill, for example, 'student advisor', as shown in Figure 2. An assistant is a chatbot that helps connect user conversation flow and search for an answer from the knowledge database.

Step 3. Design the conversation flow: Use the Watson Assistant tooling to design the conversation flow by creating nodes and defining the dialogue structure. Each node represents a step in the conversation, including user inputs and chatbot responses.